TURNING POINT

PRACTICAL RIFLES

In 1844, Captain Claude-Etienne Minié, a French military officer, developed a bullet that revolutionized firearms, making the rifle as simple to load as the common musket and increasing its firepower. Soon nearly every soldier in every nation had in his hands for the first time a weapon of almost undreamed of power, range, and accuracy. The first use of rifles on a large scale was in the Crimean War (1853–56), and it was there that the modern sniper emerged. A few years later, the use of rifles on an even larger scale helped make the American Civil War (1861–65) the deadliest in the country's history. In a short span of time, the "Minnie ball" bullet had dramatically transformed warfare.

▲ **MINIÉ BULLET**
Featuring a cavity in the base equipped with an iron cup, the original Minié bullets were plain, and tapered from base to point. Later versions, such as this one, had a cylindrical portion and grooves that were greased to lubricate the barrel, making it easier to clean. The bullet shown here is the American "Minnie ball."

The problem with rifles in the days of muzzle-loading had always been loading a ball that fit tightly enough to engage the rifling (see p.28). With a musket, the lead ball was a loose fit. With a rifle, the ball was wrapped in a patch made from greased paper or thin linen, which could be forced into the rifling grooves. After firing, gunpowder would leave thick residues in the grooves. The problematic process of loading rifles thus became even more difficult, and British riflemen in the Napoleonic Wars were issued with mallets to drive the ball down the bore after many shots had been fired.

BEFORE

Smoothbore muskets fired lead balls that were loose-fitting and might have been accurate only for an aimed shot of up to 50 yards (46m). They were more effective when used for volley-fire by ranks of men firing together, but beyond 300 yards (270m), an opponent could consider himself fairly safe, especially if moving.

- **A ROUND MUSKET BALL**, such as one made of lead, was a loose fit in the gun's bore. When fired, it would ricochet off the wall of the bore, its final direction depending upon the last point of contact.

LEAD MUSKET BALL

- **A LINEN OR PAPER PATCH** enveloping the round ball was an improvement. The ball would grip the grooves in the rifled barrel, making it spin and travel fairly accurately in flight. However, it was difficult to load.

- **THE BRUNSWICK BALL** was an example of a bullet designed to overcome existing problems. It was made to match the rifling and theoretically slide into the bore. The ball had a raised belt that fit into the two, deep rifling grooves in the Brunswick rifle. Brunswick balls could be damaged or deformed if knocked together in a pouch. Trying to align them correctly in the heat of battle also made loading difficult.

BRUNSWICK BALL

EARLY RIFLE SOLUTIONS

One route to overcoming this problem resulted in various breech-loading systems, some more successful than others. A famous example of a breech-loader was the Ferguson rifle. However, it was expensive to make and despite its superior design, only 100 units were manufactured. Other methods of loading used projectiles preformed to match the rifling. Loading rifles, however, continued to be difficult. Often, the force required to ram the ball down the bore was great enough to render the shooter's hands unsteady for accurate firing.

British officer John Jacob's rifles used four deep grooves and bullets with ribs to match. English engineer Sir Joseph Whitworth's rifle had spiral, hexagonal bores and used bullets made appropriately. Both were accurate and Whitworth's rifles were prized by sharpshooters in the American Civil War. However, they were too complex for general issue.

THE MINIÉ REVOLUTION

The solution to these problems lay in a simple bullet devised by Minié, based on his modification of a bullet created a few years earlier by fellow Frenchman Captain Henri-Gustave Delvigne. This new bullet could work with any conventional rifle. It could slide easily down the bore of a gun and at the instant of explosion, an iron cup in the bullet's base was driven into the cavity inside it, expanding the skirt of the bullet to grip the rifling grooves.

The muzzle-loading rifle evolved to become more effectual, and gradually warfare was transformed. Where once infantry could be safe beyond a distance of 300 yards (270m) from an

▸ **USING MINIÉ BULLETS**
At Fredericksburg, Virginia, in 1862, during the Civil War, the Union Army (seen here) and the Confederate defenders (entrenched outside the city) battled for weeks, many using rifles with Minié bullets.